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(54) BASEBALL BATTING TRAINING EQUIPMENT

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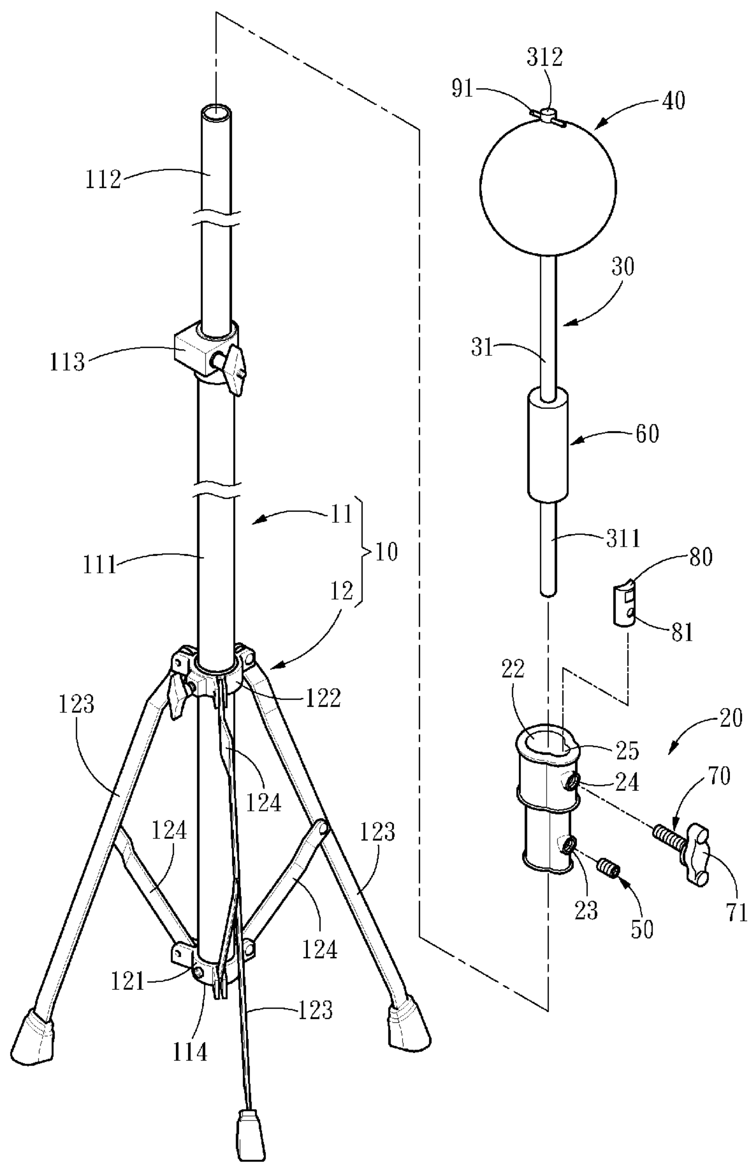
(57) **ABSTRACT**

A baseball batting training equipment, includes a tripod, a mounting base, a connecting rod, and a ball to be hit. The tripod includes a vertical tube and a support structure secured to the vertical tube to support the vertical tube in a standing position, the mounting base is fixed to the vertical tube, the connecting rod includes a rod body made of an elastic material, one end of the rod body fixed to the mounting base, and the other end of the rod body extending upward for a specified distance, the ball to be hit is fixed to the other end of the rod body; accordingly, the ball to be hit is provided for a batter to practice hand-eye coordination and is reset by elasticity of the rod body, which can meet the requirements of use such as without picking up hit balls, small space occupation, and continuous practice.



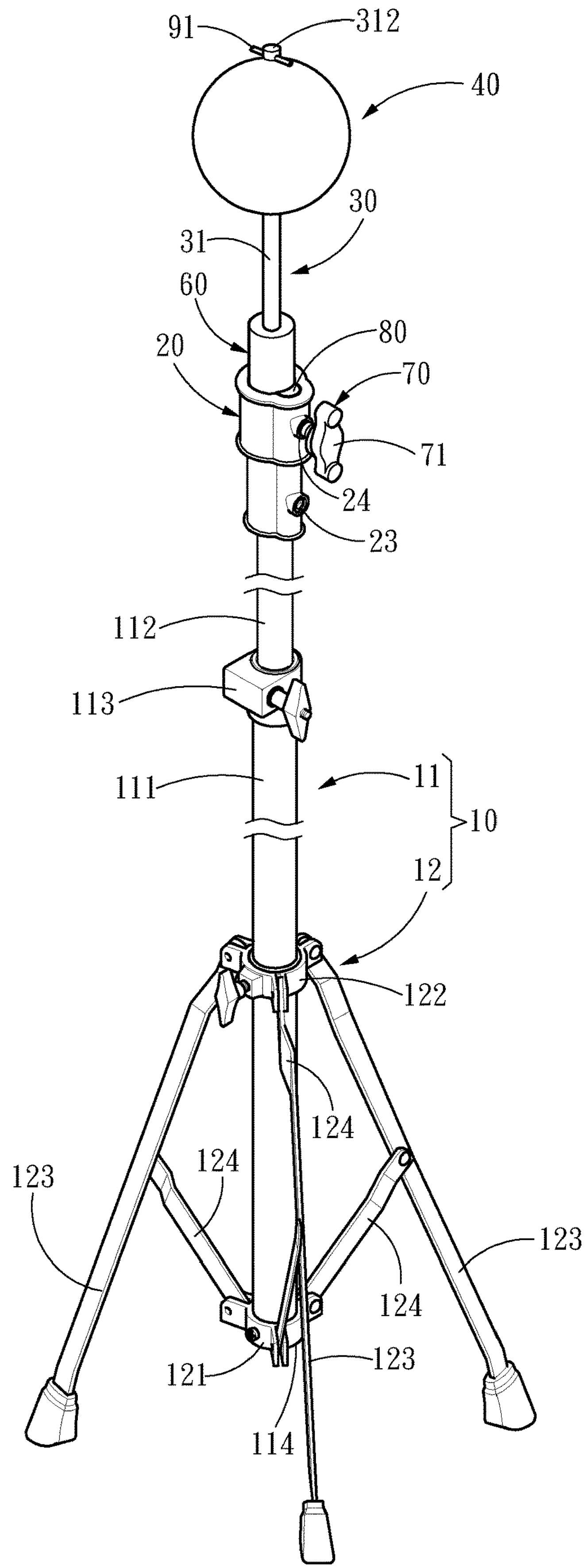

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Fig. 1

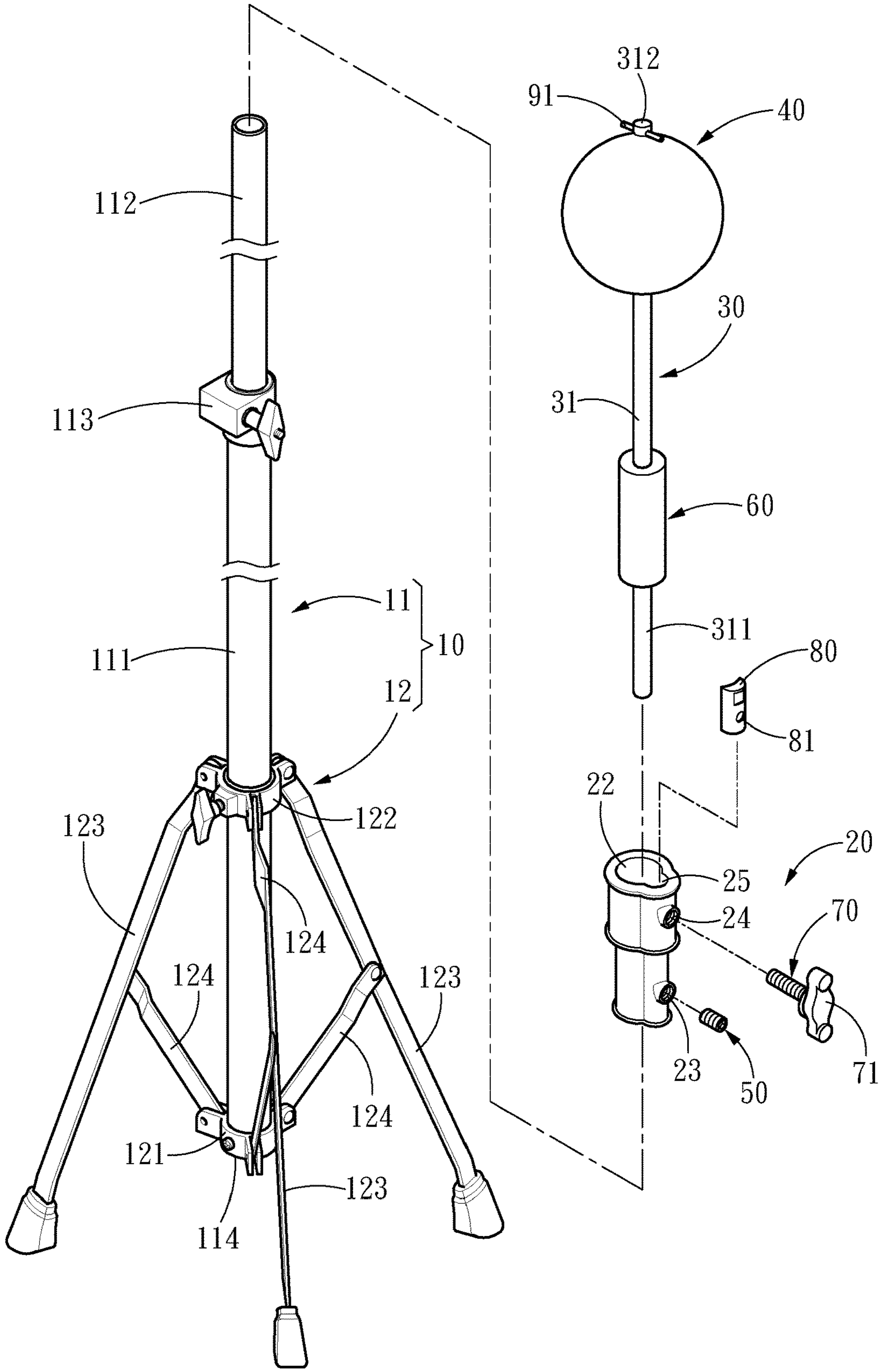
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Fig. 2

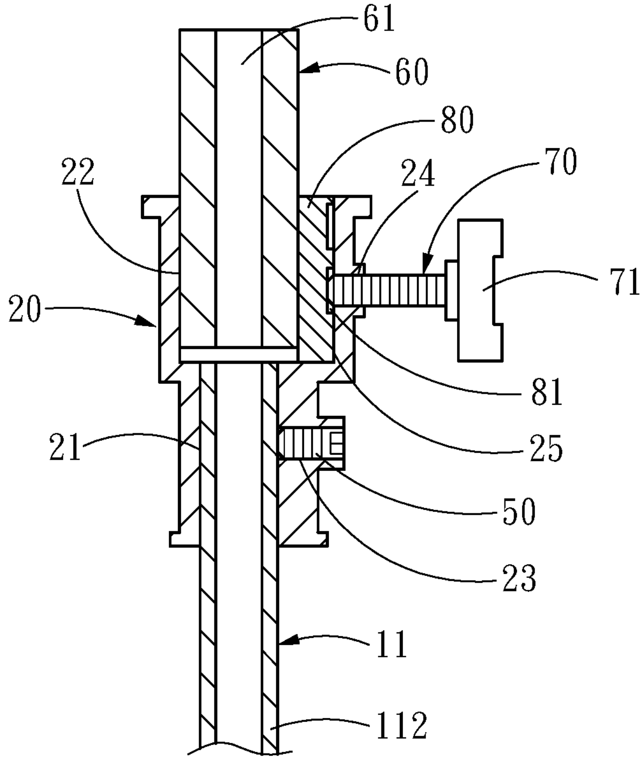

Fig. 3

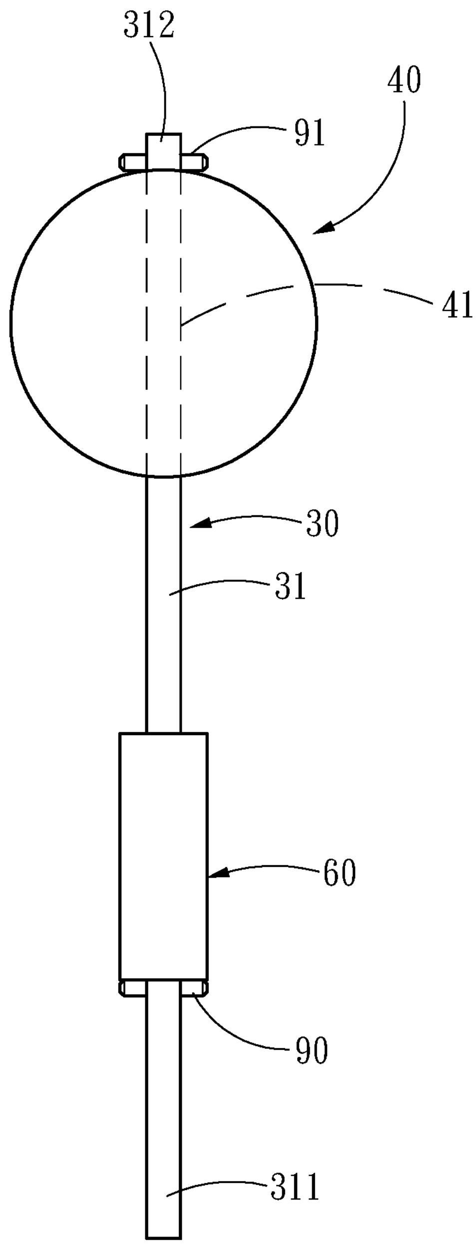

Fig. 4

BASEBALL BATTING TRAINING EQUIPMENT

FIELD OF THE INVENTION

[0001] The present invention relates to baseball training aids and, in particular, to a baseball batting training equipment.

BACKGROUND OF THE INVENTION

[0002] As disclosed in U.S. Pat. No. 11,511,175 B2, the baseball pitching machine can throw baseballs of different types, positions, and speeds for batters to practice baseball hitting. The equipment of the baseball pitching machine is expensive, also requires a protective net installed to prevent the baseball hit from endangering other people. Consequently, the cost is higher, and generally for advanced batters use.

[0003] For a batter, practicing hand-eye coordination is the first step of batting practice, therefore, U.S. Pat. No. 7,479,074 B1 discloses a batting practice device for placing a baseball on a pole, which can be used by a batter to practice hand-eye coordination, however, the baseball needs to be picked up when the baseball is batted out, and the batter needs to place the baseball on the pole again after the baseball is batted out, therefore, the efficiency of the practice is not satisfactory. For safety reasons, it is also necessary to install a protective net. Therefore, U.S. Pat. No. 7,980,966 B2 disclosed that a ball will be restored to its original position automatically by the constraint of the hanging rope when it is hit out without the need to install a protective net, which can increase convenience and reduce the cost of use. However, when the baseball is hit out, due to the constraint of the hanging rope, the baseball will be pulled back quickly and there is a risk of the batter being hit, so there is still room for improvement in use.

SUMMARY OF THE INVENTION

[0004] The main object of the present invention is to disclose a baseball batting training equipment that meets the need for practicing hand-eye coordination in batting.

[0005] In order to achieve the above object, the present invention provides a baseball batting training equipment, including a tripod, a mounting base, a connecting rod, and a ball to be hit. The tripod includes a vertical tube and a support structure secured to the vertical tube to support the vertical tube in a standing position, the mounting base is fixed to the vertical tube, the connecting rod includes a rod body made of an elastic material, one end of the rod body fixed to the mounting base, and the other end of the rod body extending upward for a specified distance, the ball to be hit is fixed to the other end of the rod body.

[0006] Accordingly, the ball to be hit is fixed to the mounting base through the connecting rod and separated by a specified distance, the rod body is made of elastic material, such that the ball to be hit of the present invention is provided for the batter to do batting practice for training hand-eye coordination, and the rod body can absorb the impact of the batting and reset by the elasticity thereof, which can meet the requirements of use such as without picking up hit balls, small space occupation, and continuous practice.

BRIEF DESCRIPTION OF THE DRAWING

[0007] FIG. **1** is a schematic view of the three-dimensional structure of the present invention.

[0008] FIG. **2** is an exploded diagram of the present invention.

[0009] FIG. **3** is a cross section view of a portion of the present invention.

[0010] FIG. **4** is a schematic view of another portion of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0011] The technical contents of the present invention are described as follows with reference to the accompanying drawings.

[0012] Please refer to FIG. **1** and FIG. **2**, the present invention provides a baseball batting training equipment, including a tripod **10**, a mounting base **20**, a connecting rod **30**, and a ball to be hit 40. The tripod **10** includes a vertical tube **11** and a support structure **12** secured to the vertical tube **11** to support the vertical tube **11** in a standing position. In one embodiment, the vertical tube **11** includes an outer tube **111**, an inner tube **112**, and a retractable lock **113**, the inner tube **112** partially penetrates the outer tube **111**, and the retractable lock **113** is configured to fix a relative position of the outer tube **111** and the inner tube **112**, such that a length of the vertical tube **11** can be adjusted by changing the relative position of the outer tube **111** and the inner tube **112**.

[0013] In one embodiment, the support structure **12** includes a fixing ring **121**, an adjustment fixture **122**, three support rods **123**, and three linking rods **124**, the fixing ring **121** is fixed to a bottom end **114** of the vertical tube **11**, the adjustment fixture **122** is locked to the vertical tube **11**, one end of each of the three support rods is connected to the adjustment fixture **122**, one end of each of the three linking rods **124** is connected to the fixing ring **121**, and the other end of each of the three linking rods **124** is connected to one of three support rods **123**. The adjustment fixture **122** is slidable relative to the vertical tube **11** after releasing to adjust an open angle formed by the three support rods **123**. When the open angle formed by the three support rods **123** is adjusted to a maximum angle, the vertical tube **11** can be secured and supported for standing stably; when the open angle formed by the three support rods **123** is adjusted to a minimum angle, it can be storage conveniently with a reduced volume.

[0014] Please refer to FIG. **3** and FIG. **4**, the mounting base **20** is fixed to the vertical tube **11**. In one embodiment, the present invention further includes a first locking fastener **50**, a buffer tube **60**, and a second locking fastener **70**. The buffer tube **60** includes a hollow passage **61**, and the mounting base **20** includes a first passage **21**, a second passage **22**, a first locking hole **23**, and a second locking hole **24**. The first locking hole **23** and second locking hole **24** respectively pass through the mounting base **20** from a side thereof to communicate with the first passage **21** and the second passage **22**.

[0015] The vertical tube **11** penetrates the first passage **21**, and the first locking fastener **50** is screwed through the first locking hole **23** to squeeze and fix to the vertical tube **11**. Preferably, the first locking fastener **50** is a countersunk screw. The buffer tube **60** is inserted into the second passage **22**, and the second locking fastener **70** is screwed through the second locking hole **24** to squeeze and fix to the buffer tube **60**, so that the buffer tube **60** is fixed to the mounting base **20**. Preferably, the second locking fastener **70** is provided with a screw handle **71** for a user to remove or fix the buffer tube **60** by hand, under a circumstance without tools.

[0016] In one embodiment, the present invention further includes an auxiliary fixing piece **80**. The mounting base **20** is formed with an accommodating slot **25** on a side of the second passage **22** and corresponding to the second locking hole **24**, and the auxiliary fixing piece **80** is placed into the accommodating slot **25** and is pressed by the second locking fastener **70**, so that the buffer tube **60** is squeezed and fixed by the auxiliary fixing piece **80** by contacting over a large area. To avoid a problem that the auxiliary fixing piece **80** slips from the second locking fastener **70**, the auxiliary fixing piece **80** is provided with a recess **81** for being abutted by the second locking fastener **70**.

[0017] The connecting rod **30** includes a rod body **31** made of an elastic material, and one end of the rod body **31** is fixed to the mounting base **20**. In an embodiment, one end of the rod body **31** penetrates the hollow passage **61** to form a protruding portion **311**, and the protruding portion **311** is transversely provided with a first limit bar **90**. A length of the first limit bar **90** is greater than a diameter of the hollow passage **61** and smaller than a diameter of the second passage **22**. By disposal of the first limit bar **90**, the rod body **31** is effectively in prevented from slipping off from the hollow passage **61**, so that the connecting rod **30** is stably fixed to the buffer tube **60**. In other words, the connecting rod **30** is fixed to the mounting base **20** through the buffer tube **60**. In addition, the first passage **21** may communicate with the second passage **22** to provide for the protruding portion **311** penetrating the first passage **21**.

[0018] The other end of the rod body **31** extends upward for a specified distance, and the ball to be hit 40 is fixed to the other end of the rod body **31**. In one embodiment, the ball to be hit 40 is provided with a through hole **41** penetrating thereof. The other end of the rod body **31** passes the through hole **41** and protrudes to form a projection post **312**, and the projection post **312** is transversely provided with a second limit bar **91**. A length of the second limit bar **91** is greater than a diameter of the through hole **41**. Accordingly, by disposal of the second limit bar **91**, the ball to be hit 40 is effectively prevented from slipping off from the rod body **31**, so that the ball to be hit 40 is stably fixed to the rod body **31**.

[0019] As mentioned above, the present invention has characteristics as follows:

[0020] 1. The ball to be hit is provided for a batter to practice hand-eye coordination and is reset by elasticity of the rod body, which can meet the requirements of use such as without picking up hit balls, small space occupation, and continuous practice.

[0021] 2. The buffer body is effectively fixed by a design of the auxiliary fixing piece contacting the buffer body over a large area.

[0022] 3. The protruding portion of the rod body is transversely provided with the first limit bar, which can effectively prevent the rod body from slipping off from the hollow passage.

[0023] 4. The projection post is transversely provided with a second limit bar, which can effectively prevent the ball to be hit from slipping off from the rod body.

What is claimed is:

1. A baseball batting training equipment, comprising:

a tripod, comprising a vertical tube and a support structure secured to the vertical tube to support the vertical tube in a standing position;

a mounting base, fixed to the vertical tube;

a connecting rod, comprising a rod body made of an elastic material, one end of the rod body fixed to the mounting base, and an other end of the rod body extending upward for a specified distance; and

a ball to be hit, fixed to the other end of the rod body.

2. The baseball batting training equipment according to claim **1**, further comprising a first locking fastener, a buffer tube, and a second locking fastener, the buffer tube comprising a hollow passage, and the mounting base comprising a first passage, a second passage, a first locking hole, and a second locking hole, wherein the first locking hole and second locking hole respectively pass through the mounting base from a side of the mounting base to communicate with the first passage and the second passage, the vertical tube penetrates the first passage, the first locking fastener screwed through the first locking hole to squeeze and fix the vertical tube, the one end of the rod body penetrates the hollow passage to form a protruding portion, and the protruding portion is transversely provided with a first limit bar, and a length of the first limit bar is greater than a diameter of the hollow passage and smaller than a diameter of the second passage.

3. The baseball batting training equipment according to claim **2**, wherein the first passage communicates with the second passage to provide for the protruding portion penetrating the first passage.

4. The baseball batting training equipment according to claim **2**, further comprising an auxiliary fixing piece, wherein the mounting base is formed with an accommodating slot on a side of the second passage and corresponding to the second locking hole, and the auxiliary fixing piece is placed into the accommodating slot and is pressed by the second locking fastener, so that the buffer tube is squeezed and fixed by the auxiliary fixing piece.

5. The baseball batting training equipment according to claim **4**, wherein the auxiliary fixing piece is provided with a recess for being abutted by the second locking fastener.

6. The baseball batting training equipment according to claim **2**, wherein the first locking fastener is a countersunk screw.

7. The baseball batting training equipment according to claim **2**, wherein the second locking fastener is a screw handle.

8. The baseball batting training equipment according to claim **1**, wherein the ball to be hit is provided with a through hole penetrating thereof, the other end of the rod body passes the through hole and protrudes to form a projection post, the projection post is transversely provided with a second limit bar, and a length of the second limit bar is greater than a diameter of the through hole.

9. The baseball batting training equipment according to claim **1**, wherein the support structure comprises a fixing ring, an adjustment fixture, three support rods, and three linking rods, the fixing ring is fixed to a bottom end of the vertical tube, the adjustment fixture is locked to the vertical tube, one end of each of the three support rods is connected to the adjustment fixture, one end of each of the three linking rods is connected to the fixing ring, and an other end of each of the three linking rods is connected to one of the three support rods.

10. The baseball batting training equipment according to claim **1**, wherein the vertical tube comprises an outer tube, an inner tube partially penetrating the outer tube, and a retractable lock configured to fix to a relative position of the outer tube and the inner tube.

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